

EMPLOYEE MANAGEMENT AND ATTENDANCE TRACKER

Create Database

Firstly, Created the database for the project Emp_Project



The screenshot shows a SQL query window with the following commands:

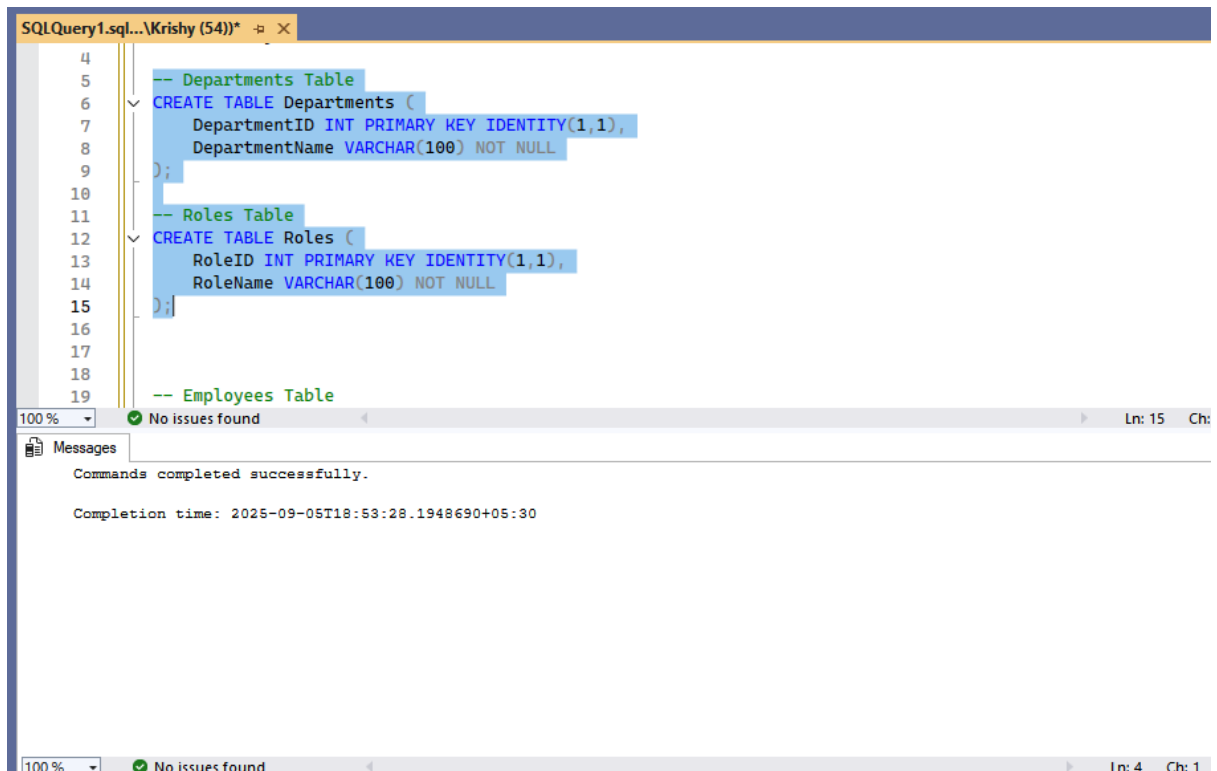
```
1 create database EMP_Project
2
3 use EMP_Project
```

The status bar indicates "No issues found". The Messages pane shows:

```
Commands completed successfully.
Completion time: 2025-09-05T18:49:24.7479558+05:30
```

Creating Table

Table definitions for Departments and Roles



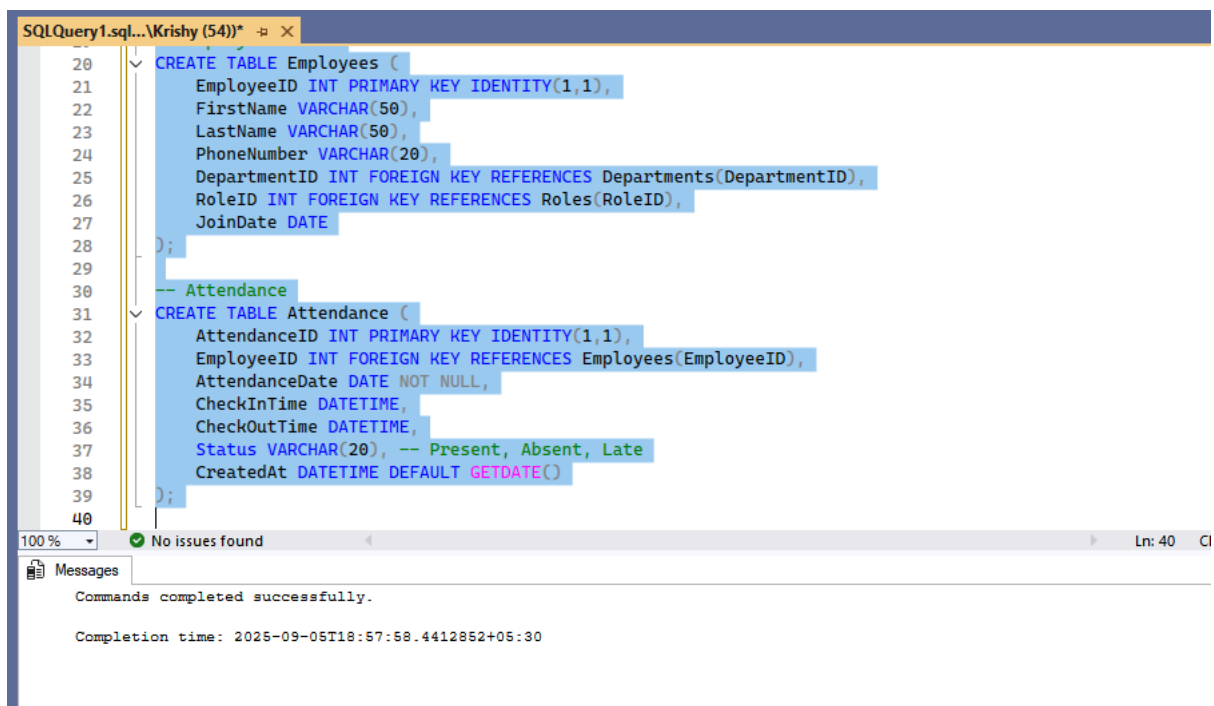
The screenshot shows a SQL query window with the following commands:

```
4
5 -- Departments Table
6 CREATE TABLE Departments (
7     DepartmentID INT PRIMARY KEY IDENTITY(1,1),
8     DepartmentName VARCHAR(100) NOT NULL
9 );
10
11 -- Roles Table
12 CREATE TABLE Roles (
13     RoleID INT PRIMARY KEY IDENTITY(1,1),
14     RoleName VARCHAR(100) NOT NULL
15 );
16
17
18
19 -- Employees Table
```

The status bar indicates "No issues found". The Messages pane shows:

```
Commands completed successfully.
Completion time: 2025-09-05T18:53:28.1948690+05:30
```

Table definitions for Employee and Attendance



```
SQLQuery1.sql... \Krishy (54))*  
20 CREATE TABLE Employees (  
21     EmployeeID INT PRIMARY KEY IDENTITY(1,1),  
22     FirstName VARCHAR(50),  
23     LastName VARCHAR(50),  
24     PhoneNumber VARCHAR(20),  
25     DepartmentID INT FOREIGN KEY REFERENCES Departments(DepartmentID),  
26     RoleID INT FOREIGN KEY REFERENCES Roles(RoleID),  
27     JoinDate DATE  
28 );  
29  
30 -- Attendance  
31 CREATE TABLE Attendance (  
32     AttendanceID INT PRIMARY KEY IDENTITY(1,1),  
33     EmployeeID INT FOREIGN KEY REFERENCES Employees(EmployeeID),  
34     AttendanceDate DATE NOT NULL,  
35     CheckInTime DATETIME,  
36     CheckOutTime DATETIME,  
37     Status VARCHAR(20), -- Present, Absent, Late  
38     CreatedAt DATETIME DEFAULT GETDATE()  
39 );  
40
```

100 % No issues found Ln: 40 CI

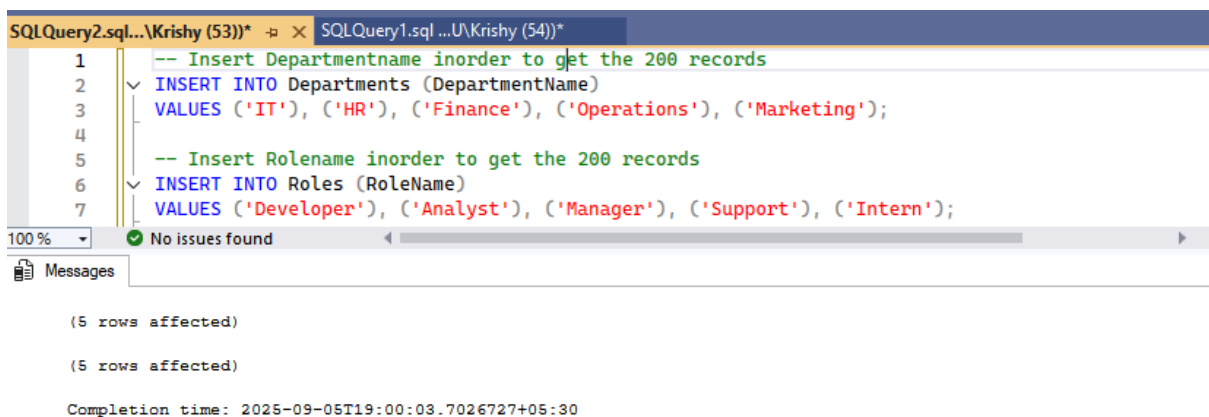
Messages

Commands completed successfully.

Completion time: 2025-09-05T18:57:58.4412852+05:30

Inserting Records

Inserted records into department and roles tables only their names are inserted. Id's are auto populated as we give identity(1,1) – it will auto increment values with Primary key constraints.



```
SQLQuery2.sql... \Krishy (53))* SQLQuery1.sql... \Krishy (54))*  
1 -- Insert Departmentname inorder to get the 200 records  
2 INSERT INTO Departments (DepartmentName)  
3 VALUES ('IT'), ('HR'), ('Finance'), ('Operations'), ('Marketing');  
4  
5 -- Insert Rolename inorder to get the 200 records  
6 INSERT INTO Roles (RoleName)  
7 VALUES ('Developer'), ('Analyst'), ('Manager'), ('Support'), ('Intern');
```

100 % No issues found

Messages

(5 rows affected)

(5 rows affected)

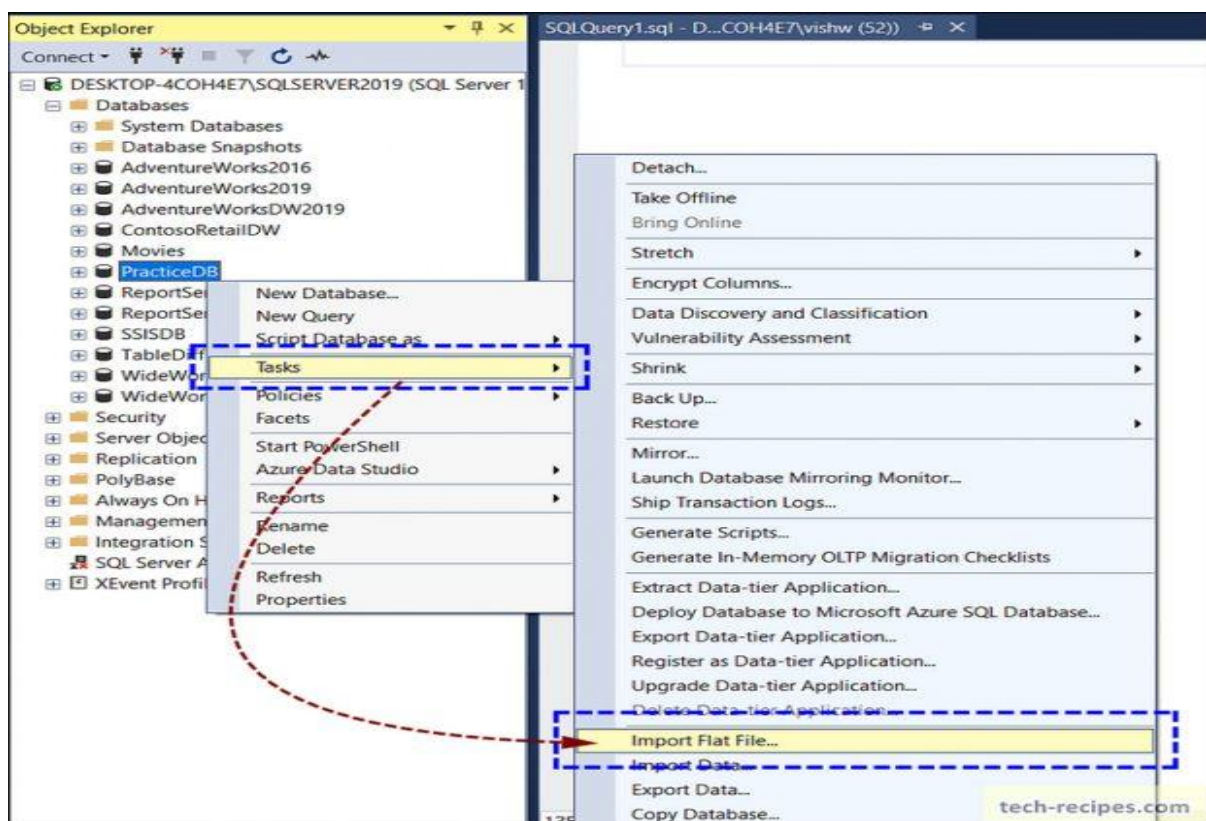
Completion time: 2025-09-05T19:00:03.7026727+05:30

Inserting 200 records in Employee and Departments Table

Have got the dummy excel from google with same table and have incorporated into SQL.

For doing so, below are the steps which I followed...

- In SQL Server Management Studio, right-click on the database where the tables are located.
- Go to Tasks -> Import flat files
- Choose path and give name to that.
- Give next and click ok.
- It give Success state after the insertion is done.



Import Flat File 'EMP_Project'

Specify Input File

Introduction
Specify Input File
Preview Data
Modify Columns
Summary
Results

[Help](#)

Specify Input File
This operation will create a table from your input file.

Location of file to be imported
 [Browse...](#)

New table name:

Table schema:

Now you can see the data are inserted.

SQLQuery3.sql... \Krishy (65))* SQLQuery2.sql ...U\Krishy (53))* SQLQuery1.sql ...U\Krishy (54))*

```
1 select * from Attendance
```

100 % No issues found Ln: 1 Ch: 25 TABS CRLF

	AttendanceID	EmployeeID	AttendanceDate	CheckInTime	CheckOutTime	Status	CreatedAt
1	1	21	2025-07-08	2025-07-08 10:01:00.0000000	2025-07-08 17:51:00.0000000	Late	2025-09-05 13:52:05.0000000
2	2	106	2025-07-19	2025-07-19 10:34:00.0000000	2025-07-19 17:36:00.0000000	Late	2025-09-05 13:52:05.0000000
3	3	24	2025-07-08	2025-07-08 09:25:00.0000000	2025-07-08 18:17:00.0000000	Present	2025-09-05 13:52:05.0000000
4	4	90	2025-07-16	2025-07-16 09:40:00.0000000	2025-07-16 18:10:00.0000000	Present	2025-09-05 13:52:05.0000000
5	5	114	2025-07-08	2025-07-08 09:32:00.0000000	2025-07-08 17:33:00.0000000	Present	2025-09-05 13:52:05.0000000
6	6	131	2025-08-31	2025-08-31 10:52:00.0000000	2025-08-31 17:57:00.0000000	Late	2025-09-05 13:52:05.0000000
7	7	56	2025-08-15	2025-08-15 09:22:00.0000000	2025-08-15 18:06:00.0000000	Present	2025-09-05 13:52:05.0000000
8	8	56	2025-07-27	2025-07-27 10:22:00.0000000	2025-07-27 17:33:00.0000000	Late	2025-09-05 13:52:05.0000000
9	9	52	2025-07-15	2025-07-15 09:05:00.0000000	2025-07-15 17:50:00.0000000	Present	2025-09-05 13:52:05.0000000
10	10	126	2025-08-11	NULL	NULL	Absent	2025-09-05 13:52:05.0000000
11	11	68	2025-08-17	2025-08-17 10:42:00.0000000	2025-08-17 18:13:00.0000000	Late	2025-09-05 13:52:05.0000000

Query executed successfully. localhost (16.0 RTM) SIVU\Krishy (65) EMP_Project 00:00:00 200 rows

SQLQuery3.sql... \Krishy (65))* SQLQuery2.sql ...U\Krishy (53))* SQLQuery1.sql ...U\Krishy (54))*

```
1 select * from Employees
```

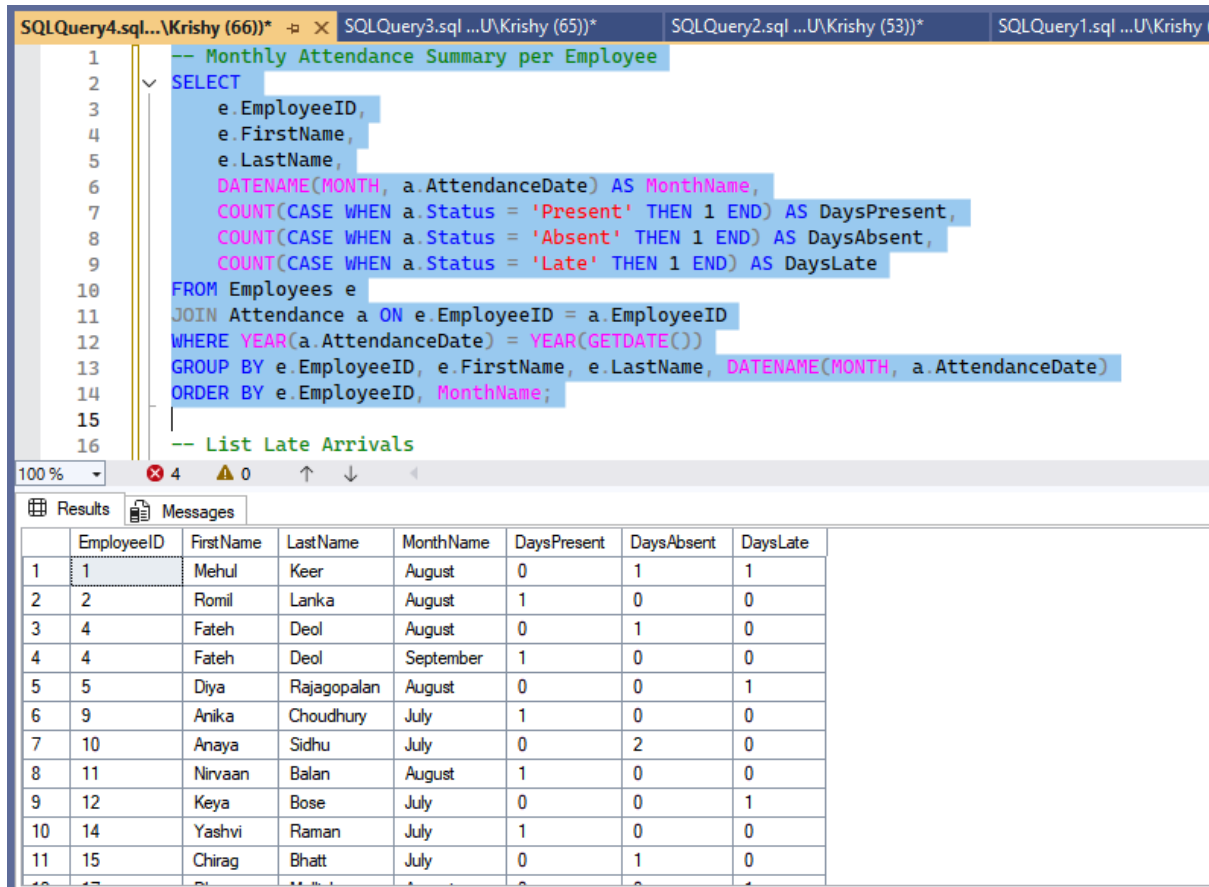
100 % No issues found Ln: 1 Ch: 24 TABS CRLF

	EmployeeID	FirstName	LastName	PhoneNumber	DepartmentID	RoleID	JoinDate
1	1	Mehul	Keer	8672645528	3	4	2025-07-18
2	2	Romil	Lanka	3872685352	2	4	2022-11-06
3	3	Ojas	Choudhary	7292957608	5	4	2025-01-13
4	4	Fateh	Deol	1192173642	1	2	2024-12-07
5	5	Diya	Rajagopalan	5233889529	3	4	2023-02-28
6	6	Yakshit	Tara	5028837318	1	2	2025-02-03
7	7	Jayesh	Gandhi	1595638548	4	5	2025-01-02
8	8	Myra	Majumdar	2279794805	4	1	2025-08-29
9	9	Anika	Choudhury	90957200	1	3	2024-06-16
10	10	Anaya	Sidhu	7804231724	5	2	2025-04-10
11	11	Nirvaan	Balan	8706892307	4	4	2025-05-25

Query executed successfully. localhost (16.0 RTM) SIVU\Krishy (65) EMP_Project 00:00:00 200 rows

Monthly attendance

Here you see Monthly attendance for each students...



The screenshot shows a SQL Server Enterprise Manager window with a query editor and a results pane. The query editor contains a SQL query for monthly attendance summary. The results pane shows a table with 8 columns: EmployeeID, FirstName, LastName, MonthName, DaysPresent, DaysAbsent, and DaysLate. The table contains 11 rows of data.

```
1 -- Monthly Attendance Summary per Employee
2 SELECT
3     e.EmployeeID,
4     e.FirstName,
5     e.LastName,
6     DATENAME(MONTH, a.AttendanceDate) AS MonthName,
7     COUNT(CASE WHEN a.Status = 'Present' THEN 1 END) AS DaysPresent,
8     COUNT(CASE WHEN a.Status = 'Absent' THEN 1 END) AS DaysAbsent,
9     COUNT(CASE WHEN a.Status = 'Late' THEN 1 END) AS DaysLate
10 FROM Employees e
11 JOIN Attendance a ON e.EmployeeID = a.EmployeeID
12 WHERE YEAR(a.AttendanceDate) = YEAR(GETDATE())
13 GROUP BY e.EmployeeID, e.FirstName, e.LastName, DATENAME(MONTH, a.AttendanceDate)
14 ORDER BY e.EmployeeID, MonthName;
15
16 -- List Late Arrivals
```

	EmployeeID	FirstName	LastName	MonthName	DaysPresent	DaysAbsent	DaysLate
1	1	Mehul	Keer	August	0	1	1
2	2	Romil	Lanka	August	1	0	0
3	4	Fateh	Deol	August	0	1	0
4	4	Fateh	Deol	September	1	0	0
5	5	Diya	Rajagopalan	August	0	0	1
6	9	Anika	Choudhury	July	1	0	0
7	10	Anaya	Sidhu	July	0	2	0
8	11	Nirvaan	Balan	August	1	0	0
9	12	Keya	Bose	July	0	0	1
10	14	Yashvi	Raman	July	1	0	0
11	15	Chirag	Bhatt	July	0	1	0

Late arrivals

Here you see Late arrivals for each students...

SQLQuery4.sql... \Krishy (66))*					
<pre> 14 ORDER BY e.EmployeeID, MonthName; 15 16 -- List Late Arrivals 17 SELECT 18 e.EmployeeID, 19 e.FirstName, 20 e.LastName, 21 a.AttendanceDate, 22 a.CheckInTime 23 FROM Employees e 24 JOIN Attendance a ON e.EmployeeID = a.EmployeeID 25 WHERE a.Status = 'Late' 26 ORDER BY a.AttendanceDate DESC; 27 </pre>					
100 % 4 0 ↑ ↓					
Results					
	EmployeeID	FirstName	LastName	AttendanceDate	CheckInTime
1	113	Amaan	Edwin	2025-09-02	2025-09-02 10:52:00.0000000
2	156	Rasha	Kapoor	2025-09-02	2025-09-02 10:07:00.0000000
3	182	Nitara	Sankaran	2025-08-31	2025-08-31 10:10:00.0000000
4	131	Vidur	Goyal	2025-08-31	2025-08-31 10:52:00.0000000
5	72	Kaan	Kari	2025-08-30	2025-08-30 10:28:00.0000000
6	151	Jiya	Magar	2025-08-30	2025-08-30 10:59:00.0000000
7	17	Dhruv	Mallick	2025-08-27	2025-08-27 10:06:00.0000000
8	91	Himmat	Sandhu	2025-08-25	2025-08-25 10:45:00.0000000
9	112	Faiyaz	Deep	2025-08-22	2025-08-22 10:01:00.0000000
10	159	Ayesha	Rege	2025-08-21	2025-08-21 10:46:00.0000000
11	62	Eva	Chacko	2025-08-20	2025-08-20 10:13:00.0000000

Triggers

Triggers are created for updating the records.

SQLQuery5.sql... \Krishy (65))*			
<pre> 1 CREATE TRIGGER trg_SetAttendanceStatus 2 ON Attendance 3 AFTER INSERT 4 AS 5 BEGIN 6 UPDATE a 7 SET 8 a.CreatedAt = GETDATE(), 9 a.Status = CASE 10 WHEN i.CheckInTime IS NULL THEN 'Absent' 11 WHEN CAST(i.CheckInTime AS time) > '09:30:00' THEN 'Late' 12 ELSE 'Present' 13 END 14 FROM Attendance a 15 INNER JOIN inserted i ON a.AttendanceID = i.AttendanceID; 16 END; </pre>			
100 % No issues found Ln: 16			
Messages			
Commands completed successfully. Completion time: 2025-09-05T22:27:14.0971189+05:30			

You can see the below records are updated.

SQLQuery5.sql...\\Krishy (65))* SQLQuery4.sql ...U\\Krishy (66))* SQLQuery2.sql ...U\\Krishy (53))* SQLQuery1.sql ...U\\Kr

9 UPDATE A

10 SET

11 A.CreatedAt = GETDATE(),

12 A.Status = CASE

13 WHEN A.CheckInTime IS NULL THEN 'Absent'

14 WHEN CAST(A.CheckInTime AS TIME) > '09:30:00' THEN 'Late'

15 ELSE 'Present'

16 END

17 FROM Attendance A

18 INNER JOIN Inserted I ON A.AttendanceID = I.AttendanceID;

19 END;

20 GO

21

22

23 -- Employee Absent

24 INSERT INTO Attendance (AttendanceID, EmployeeID, AttendanceDate, CheckInTime, CheckOut

25 VALUES (205, 3, '2025-08-09', NULL, NULL, 'Absent', '2025-09-05 18:30:00');

26

27

100 %

2 0

Messages

(2 rows affected)

(2 rows affected)

(1 row affected)

Completion time: 2025-09-05T22:55:57.7680419+05:30

100 %

No issues found

--

21

22

23 -- Employee Absent

24 INSERT INTO Attendance (AttendanceID, EmployeeID, AttendanceDate, CheckInTime, CheckOutTime, Status, CreatedAt)

25 VALUES (201, 5, '2025-09-05', NULL, NULL, 'Absent', '2025-09-05 18:30:00');

26

27

28 SELECT AttendanceID, EmployeeID, AttendanceDate, CheckInTime, CheckOutTime, Status, CreatedAt

29 FROM Attendance

WHERE AttendanceDate = '2025-09-05';

100 %

2 0

Ln: 29 Ch: 37 SPC

Results Messages

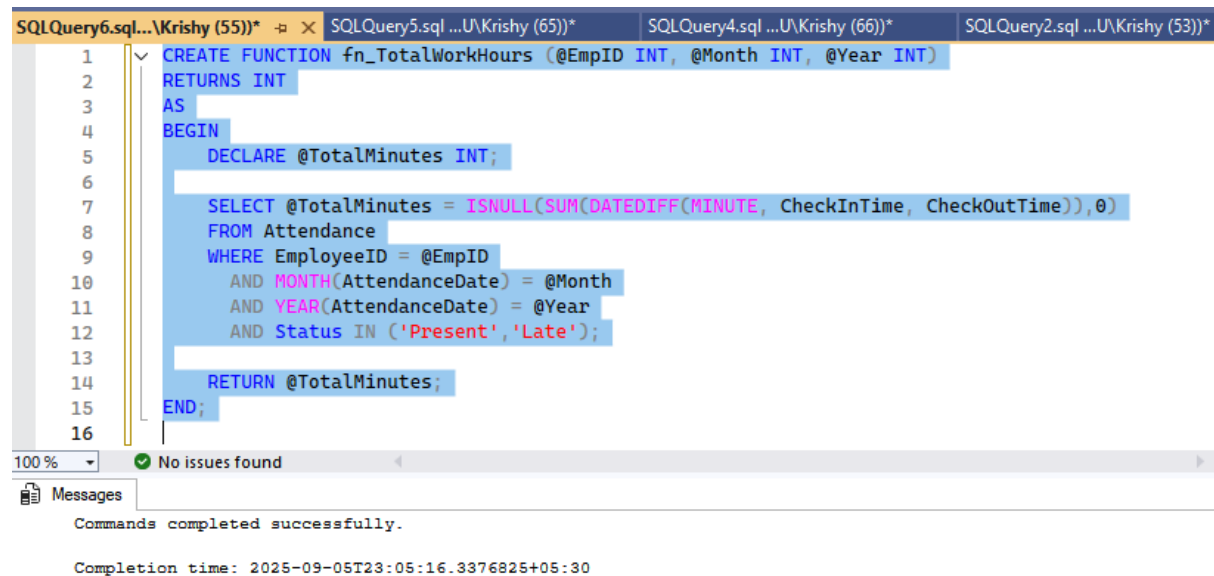
	AttendanceID	EmployeeID	AttendanceDate	CheckInTime	CheckOutTime	Status	CreatedAt
1	201	1	2025-09-05	2025-09-05 09:05:00.0000000	2025-09-05 17:30:00.0000000	Present	2025-09-05 23:02:11.4400000
2	202	4	2025-09-05	2025-09-05 09:05:00.0000000	2025-09-05 17:30:00.0000000	Present	2025-09-05 22:48:40.9300000
3	201	3	2025-09-05	NULL	NULL	Absent	2025-09-05 23:02:11.4400000
4	205	5	2025-09-05	NULL	NULL	Absent	2025-09-05 23:01:07.1000000
5	201	5	2025-09-05	NULL	NULL	Absent	2025-09-05 23:02:11.4400000

Query executed successfully

localhost (16.0 RTM) \\SIVIA\\Krishy (65) \\EMP_Project 00:00:00 5

Functions

Here I created the function to calculate the total work hours



```
SQLQuery6.sql... \Krishy (55))*  SQLQuery5.sql ...U\Krishy (65))*  SQLQuery4.sql ...U\Krishy (66))*  SQLQuery2.sql ...U\Krishy (53))*
1 CREATE FUNCTION fn_TotalWorkHours (@EmpID INT, @Month INT, @Year INT)
2 RETURNS INT
3 AS
4 BEGIN
5     DECLARE @TotalMinutes INT;
6
7     SELECT @TotalMinutes = ISNULL(SUM(DATEDIFF(MINUTE, CheckInTime, CheckOutTime)), 0)
8     FROM Attendance
9     WHERE EmployeeID = @EmpID
10        AND MONTH(AttendanceDate) = @Month
11        AND YEAR(AttendanceDate) = @Year
12        AND Status IN ('Present', 'Late');
13
14     RETURN @TotalMinutes;
15 END;
16
```

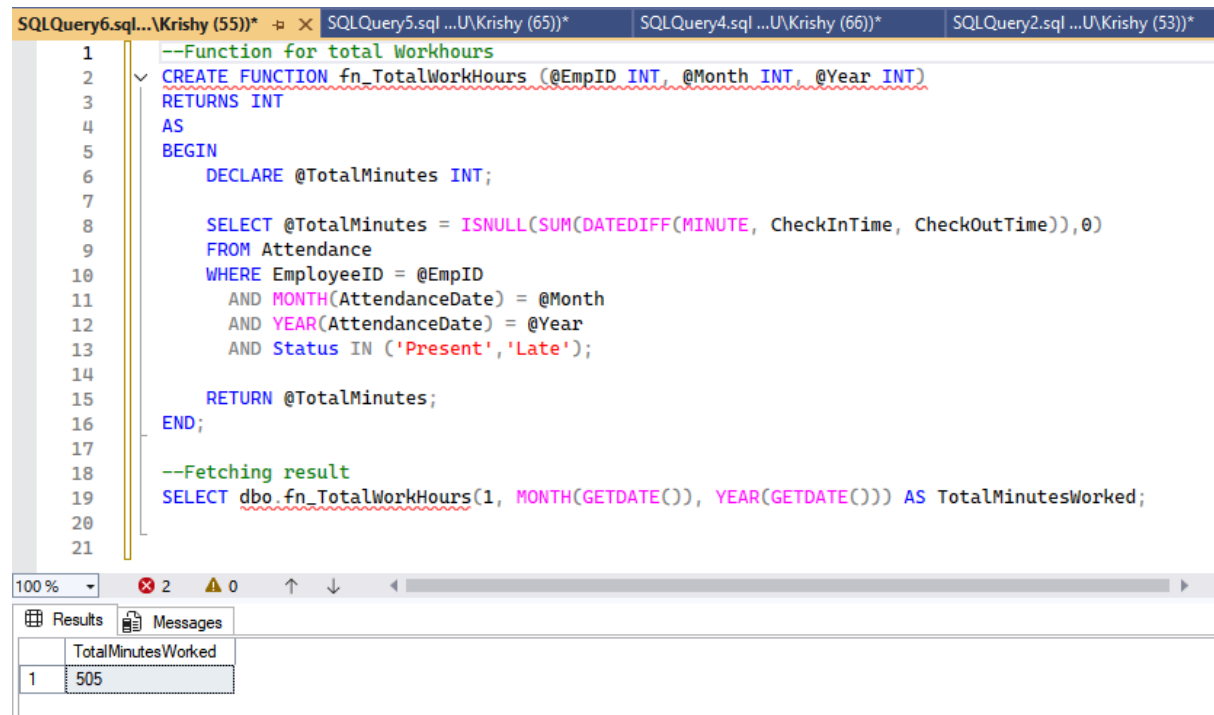
100 % No issues found

Messages

Commands completed successfully.

Completion time: 2025-09-05T23:05:16.3376825+05:30

Fetches the total hours of Employee having Id = 1.



```
SQLQuery6.sql... \Krishy (55))*  SQLQuery5.sql ...U\Krishy (65))*  SQLQuery4.sql ...U\Krishy (66))*  SQLQuery2.sql ...U\Krishy (53))*
1 --Function for total Workhours
2 CREATE FUNCTION fn_TotalWorkHours (@EmpID INT, @Month INT, @Year INT)
3 RETURNS INT
4 AS
5 BEGIN
6     DECLARE @TotalMinutes INT;
7
8     SELECT @TotalMinutes = ISNULL(SUM(DATEDIFF(MINUTE, CheckInTime, CheckOutTime)), 0)
9     FROM Attendance
10    WHERE EmployeeID = @EmpID
11        AND MONTH(AttendanceDate) = @Month
12        AND YEAR(AttendanceDate) = @Year
13        AND Status IN ('Present', 'Late');
14
15     RETURN @TotalMinutes;
16 END;
17
18 --Fetching result
19 SELECT dbo.fn_TotalWorkHours(1, MONTH(GETDATE()), YEAR(GETDATE())) AS TotalMinutesWorked;
20
21
```

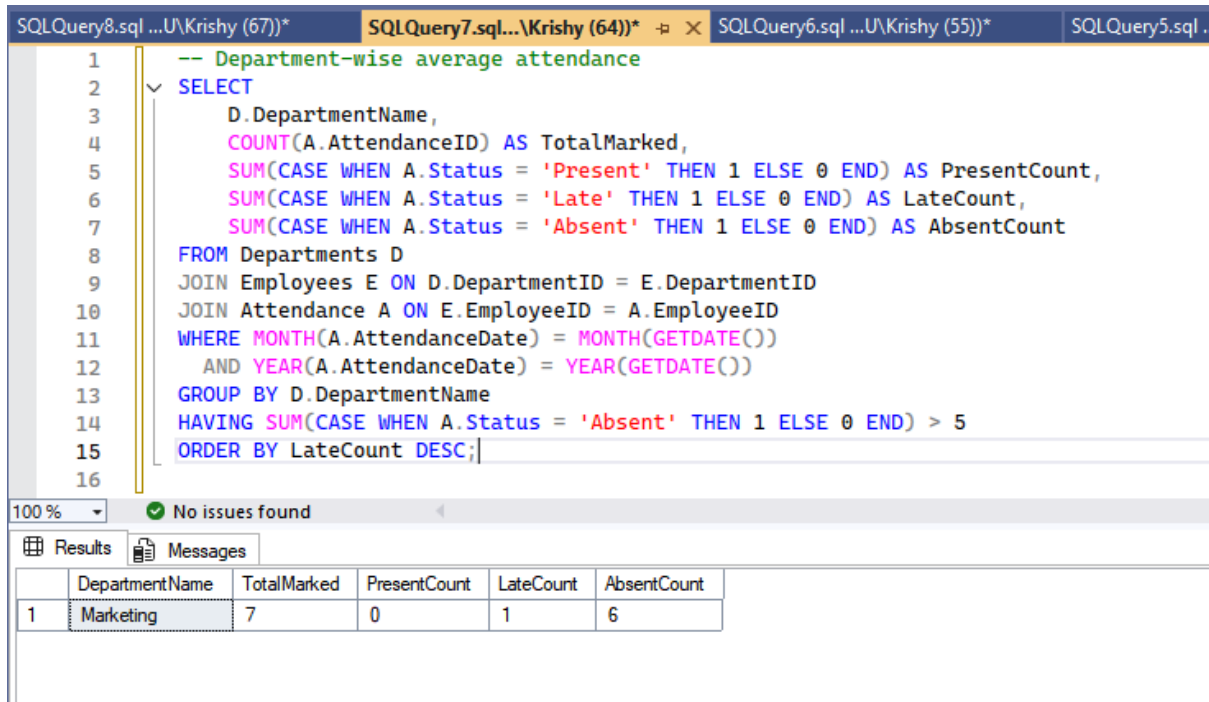
100 % 2 0

Results Messages

	TotalMinutesWorked
1	505

Groupby & Having

Have created the report for Department wise average attendance



The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows a SQL query for "Department-wise average attendance". The query uses GROUP BY and HAVING clauses to filter departments based on the number of absent employees. The bottom pane shows the results of the query, which includes a single row for the Marketing department.

```
1  -- Department-wise average attendance
2  SELECT
3      D.DepartmentName,
4      COUNT(A.AttendanceID) AS TotalMarked,
5      SUM(CASE WHEN A.Status = 'Present' THEN 1 ELSE 0 END) AS PresentCount,
6      SUM(CASE WHEN A.Status = 'Late' THEN 1 ELSE 0 END) AS LateCount,
7      SUM(CASE WHEN A.Status = 'Absent' THEN 1 ELSE 0 END) AS AbsentCount
8  FROM Departments D
9  JOIN Employees E ON D.DepartmentID = E.DepartmentID
10 JOIN Attendance A ON E.EmployeeID = A.EmployeeID
11 WHERE MONTH(A.AttendanceDate) = MONTH(GETDATE())
12       AND YEAR(A.AttendanceDate) = YEAR(GETDATE())
13 GROUP BY D.DepartmentName
14 HAVING SUM(CASE WHEN A.Status = 'Absent' THEN 1 ELSE 0 END) > 5
15 ORDER BY LateCount DESC;
```

100 % No issues found

Results Messages

	DepartmentName	TotalMarked	PresentCount	LateCount	AbsentCount
1	Marketing	7	0	1	6

THANK YOU!